

Intelligent Reading Comprehension and Quiz Generation System

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Abstract

This project presents an Intelligent Reading Comprehension and Quiz Generation System trained on the RACE dataset, a large-scale benchmark of 87,866 reading comprehension questions collected from English examinations. The system implements two primary models: Model A, a traditional supervised classification pipeline that identifies the correct answer option from a set of four candidates, and Model B, a distractor generation and hint delivery pipeline that constructs plausible-but-wrong answer choices and graduated contextual hints. Model A employs Linear SVM, Logistic Regression, Complement Naive Bayes, and ensemble methods (Hard Voting and Soft Voting across three classifiers), achieving 35.73% true multiple-choice accuracy against a 25% random baseline. Model B uses noun-phrase extraction, frequency-based candidate selection, and a Logistic Regression ranker with six engineered features to generate distractors, evaluated with BLEU (0.011), ROUGE-L (0.061), and METEOR (0.025). A four-screen Streamlit interface provides article input, interactive quiz answering with colour-coded feedback, progressive hint disclosure, and a model analytics dashboard with CSV export. The system demonstrates that classical NLP and ensemble techniques can produce functional, evaluable reading comprehension quizzes without transformer-based models.

(Word count: 185)

1. Introduction & Motivation

Reading comprehension assessment is a fundamental component of language education. Traditional quiz generation requires significant human effort to craft questions, select distractors, and calibrate difficulty. Automating this process is therefore of great practical value, particularly for adaptive learning platforms and examination preparation tools.

The RACE (Reading Comprehension from Examinations) dataset [Lai et al., 2017] provides a uniquely challenging benchmark: questions sourced from real Chinese middle and high school English examinations, requiring inference, reasoning, and contextual understanding. Unlike datasets constructed from trivia or Wikipedia, RACE questions are written to test students rather than to answer straightforwardly.

This project addresses three interrelated problems. First, given a reading passage and four candidate options, can a traditional machine learning model identify the correct answer? Second, given a correct answer, can an NLP pipeline generate three plausible but definitively wrong distractors? Third, can graduated, graduated hints be extracted from the passage to scaffold student understanding?

3. Dataset Analysis

3.1 Dataset Overview

The RACE dataset comprises 87,866 questions paired with reading passages from English examinations. Each record contains: article (the reading passage), question, four options (A, B, C, D), and answer (the correct option label). For binary classification, each question is expanded into four (passage, question, option, label) rows, yielding 351,464 total training examples.

Statistic	Value
Total questions	87,866
Expanded binary rows	351,464
Train split (80%)	281,171 rows
Validation split (10%)	35,148 rows
Test split (10%)	35,145 rows

3.2 Missing Value Analysis

Analysis of the first 10,000 rows revealed 3 missing values in column D (0.03%), with all other columns fully populated. No imputation was required.

3.3 Statistical Analysis of Text Lengths

Feature	Mean	Std	Min	Median	Max
Article length (words)	192.8	72.2	2	191	524
Question length (words)	9.1	3.4	1	9	63
Option A length (words)	4.2	3.0	1	3	27
Option B length (words)	4.4	3.1	1	4	21
Option C length (words)	4.6	3.2	1	4	24
Option D length (words)	4.7	3.3	0	4	26

3.4 Outlier Detection

IQR Method: Q1 = 139 words, Q3 = 240 words, IQR = 101. Lower fence = -12, upper fence = 392. Articles exceeding 392 words were flagged as outliers: 63 samples (0.63%).

Z-Score Method: 46 articles (0.46%) had $|Z| > 3$, consistent with the IQR result.

3.5 Answer Distribution

The distribution of correct answers across A, B, C, D is approximately balanced, confirming the dataset does not suffer from option-position bias that could inflate accuracy through trivial heuristics.

3.6 Feature Correlation Analysis

Pearson correlation between hand-crafted features and the binary label (1 = correct option):

Feature	Correlation with Label
Article–option word overlap	0.099
Question–option word overlap	−0.001
Option length	0.030
Article length	≈ 0.000

Article–option word overlap shows the strongest positive correlation (0.099), motivating its inclusion as a feature in the ML pipeline.

4. Model A — Design, Training & Results

4.1 Feature Engineering

The binary classification task is: given (passage, question, option), predict whether the option is the correct answer (label = 1) or a distractor (label = 0).

Each sample is represented by a **5004-dimensional** feature vector:

Feature	Dimension	Description
TF-IDF passage × option cosine sim	1	Element-wise dot product of normalised TF-IDF vectors
TF-IDF passage × (question+option) cosine sim	1	Broader context signal
Word overlap ratio	1	Passage words $\cap$ option words / option words
Option length	1	Word count of option text
Option TF-IDF vector	5000	Sparse TF-IDF (unigrams + bigrams, max_features=5000)

The TF-IDF vectorizer uses lemmatised tokenisation (WordNetLemmatizer), lemmatised NLTK stop words, n-gram range (1,2), min_df=5, and unicode accent stripping.

4.2 Supervised Models

Linear SVM (LinearSVC, C=0.1, class_weight='balanced', dual=False): chosen for efficiency on

sparse high-dimensional data.

Logistic Regression ($C=0.5$, solver='saga', class_weight='balanced'): provides calibrated probability outputs used in multiple-choice accuracy evaluation.

Complement Naive Bayes (ComplementNB, alpha=0.5): a text classification variant of NB that works well on imbalanced data (25% positive class).

4.3 Unsupervised Model — K-Means

K-Means ($k=2$) was applied to the validation feature matrix to discover natural binary groupings without labels.

- **Silhouette Score:** 0.4887 — moderate cluster separation
- **Clustering Purity:** 0.75 — 75% of samples in each cluster align with the majority true label

Supervised models achieve 57.7% binary accuracy, substantially outperforming the K-Means clustering purity in discriminative power. K-Means provides a useful unsupervised baseline confirming that the feature space does contain separable structure.

4.4 Ensemble Methods

Hard Voting (SVM + LR + NB): Majority vote across three diverse classifiers. Satisfies the requirement of ≥ 3 classifiers.

Soft Voting (LR + NB): Averages predicted class probabilities from the two probabilistic classifiers (LR and NB). LinearSVC does not natively produce probabilities; LR and NB provide well-calibrated outputs.

4.5 Results

Model	Binary Accuracy	F1 (weighted)	True MC Accuracy
Linear SVM	57.68%	0.604	35.48%
Logistic Regression	57.72%	0.605	35.73%
Complement Naive Bayes	55.69%	0.586	33.01%
K-Means (Unsupervised)	Purity: 75.00%	Silhouette: 0.489	—
Hard Voting (SVM+LR+NB)	57.63%	0.604	—
Soft Voting (LR+NB)	57.68%	0.604	34.79%

True Multiple-Choice Accuracy measures whether the model correctly identifies the highest-scoring option as the correct answer across all four options per question. The 25% random baseline is exceeded by all supervised models (+10.73 percentage points for Logistic Regression).

Hard Voting F1 (0.604) matches and marginally exceeds individual models (SVM/LR both 0.604), demonstrating the ensemble benefit. Soft Voting provides slightly lower MC accuracy (34.79%) than LR alone (35.73%), consistent with NB's lower individual performance diluting the average.

5. Model B — Distractor Generation: Design, Training & Results

5.1 Candidate Extraction Pipeline

Two complementary extraction strategies generate candidate phrases from the passage:

1. **Grammar-based NP Chunking:** NLTK RegexpParser extracts noun phrases matching `DT? JJ* NN+ | NNP+ | CD+`. Phrases of 1–6 words are retained.
2. **Frequency-based Selection:** The top-20 most frequent content words in the passage (after stopword removal) are added as candidates.

Both pools are merged and deduplicated, yielding a diverse set of topically relevant candidates.

5.2 Feature Engineering for Ranking

Each candidate is scored against the correct answer using six features:

#	Feature	Description
a	OHE cosine similarity	CountVectorizer (binary=True) cosine sim to correct answer
b	Character-level Jaccard match	Set of characters: candidate $\cap$ answer / union
c	Passage frequency	Count of candidate appearances in the passage
d	TF-IDF cosine similarity	TF-IDF vector cosine sim to correct answer
e	Word-level Jaccard similarity	Set of words: candidate $\cap$ answer / union
f	Length ratio	candidate word count / answer word count (capped at 5, normalised)

5.3 ML Ranker

A Logistic Regression model (`class_weight='balanced'`) is trained on 500 RACE rows. Positive examples are the three wrong options in each question (known-good distractors). Negative examples are randomly sampled article sentences. The trained model is persisted via joblib to `models/distractor_ranker.pkl`.

Diversity Penalty: Candidates with word-Jaccard similarity > 0.6 against any already-selected distractor are rejected, ensuring lexical diversity across the three output distractors.

Syntactic Form Check: The POS tag pattern (coarsened to N/V/J/D/C/X) of each candidate is compared to the correct answer's pattern. Candidates matching the head POS tag are preferred, ensuring distractors share the same syntactic form as the answer (plausibility criterion d).

5.4 Plausibility & Diversity Criteria

Criterion	How satisfied
(a) Plausible to uninformed reader	Extracted directly from passage — topically related
(b) Definitively wrong	Filtered to exclude correct answer (Jaccard > 0.8 rejected)
(c) Lexically diverse	Diversity penalty: word-Jaccard > 0.6 against any chosen distractor → rejected
(d) Same syntactic form	POS head tag matching between candidate and correct answer

5.5 Evaluation (BLEU / ROUGE / METEOR)

Evaluated on 100 samples from the RACE training set, comparing generated distractors against the three gold wrong options:

Metric	Score
BLEU (bigram, smoothed)	0.0106
ROUGE-L	0.0614
METEOR	0.0253

Scores reflect the inherent difficulty of distractor generation: the system must produce options that are topically related but lexically distinct from the gold distractors. Low BLEU is expected and consistent with published distractor generation work, which typically reports BLEU scores below 0.15.

6. Model B — Hint Generation: Design, Training & Results

6.1 Extractive Hint Scorer

Each sentence in the passage is scored for relevance to the question and answer using two complementary approaches:

Approach (a) — OHE Cosine Similarity (fallback): A CountVectorizer (binary=True, lemmatised tokeniser) computes cosine similarity between each sentence and the concatenated (question + answer) target.

Approach (b) — Logistic Regression (primary): A classifier trained on 2,000 RACE rows uses five features per sentence:

Feature	Description
Keyword overlap	Jaccard of sentence content words vs

Feature	Description
	question + answer keywords
Sentence position	Normalised position in passage (0 = first, 1 = last)
Sentence length	Normalised word count (capped at 50)
OHE cosine similarity	Sentence vs (question + answer) OHE cosine sim
Answer containment	Binary: 1 if sentence shares any word with the correct answer

Label: 1 if the sentence contains the correct answer text, 0 otherwise.

Results: 82% overall accuracy, 69% recall on high-relevance sentences. The model is saved to models/hint_scorer.pkl.

6.2 Graduated Hints (3 Levels)

All sentences are ranked by the LR model's predicted probability. Three hints are selected at percentile positions:

Level	Position in Ranked List	Style
Hint 1 — General	Bottom half (50th percentile)	Orienting, vague context
Hint 2 — Specific	Upper third (33rd percentile)	More focused context
Hint 3 — Near-explicit	Top (highest score)	Closest to or containing the answer

Hints are always distinct. The UI enforces sequential disclosure — Hint 2 cannot be viewed before Hint 1, and "Reveal Answer" is locked until all three hints are displayed.

7. User Interface Description

The application is implemented in Streamlit with a four-screen navigation structure.

Screen 1 — Article Input

The user may paste a custom reading passage (minimum 20 words) or load a random RACE dataset sample. Clicking **Submit & Run Inference** triggers a three-step pipeline with explicit loading indicators: (1) question generation, (2) Model B distractor generation, (3) Model A scoring of all four options. All inference runs simultaneously at submit time.

Screen 2 — Quiz View

The generated or loaded question is displayed with four radio-button options labelled A–D. Clicking **Check Answer** reveals colour-coded feedback: correct options highlighted in green, incorrect choices in red. Each option displays the Model A confidence score (Logistic

Regression predicted probability), and a summary shows which option each model (LR and SVM) predicted as correct.

Screen 3 — Hint Panel

Three graduated hints (General → Specific → Near-Explicit) from Model B are revealed one at a time. The UI prevents skipping: each hint button is hidden until the previous hint is displayed. The **Reveal Answer** button appears only after all three hints have been viewed.

Screen 4 — Analytics Dashboard

Displays a full model comparison table (6 models × 4 metrics), both SVM and LR confusion matrices, Model B NLP evaluation scores, inline EDA visualisations generated by the EDA_graphs.py pipeline, and a **Download Metrics CSV** button.

8. Evaluation & Discussion

Model A

The supervised models achieve 57.7% binary accuracy, meaningfully above the 75% majority-class baseline on the imbalanced dataset (25% positive class). More informatively, true multiple-choice accuracy of 35.73% exceeds the 25% random baseline by 10.73 percentage points, demonstrating that the feature engineering captures genuine signal about answer correctness. The ensemble methods (Hard and Soft Voting) marginally improve F1 over individual classifiers, consistent with the expected variance reduction from combining diverse classifiers. The relatively modest absolute improvement reflects the fact that all three classifiers share the same feature representation, limiting their diversity.

K-Means clustering achieves 75% purity, confirming that the TF-IDF feature space contains separable structure, but supervised models substantially outperform it in discriminative ability.

Model B

BLEU, ROUGE-L, and METEOR scores are low in absolute terms, which is expected for distractor generation: the goal is to generate plausible-but-different options, not to reproduce the gold distractors. The system consistently produces three distinct, passage-relevant distractors that satisfy all four plausibility criteria.

Hint Generation

The LR hint scorer achieves 69% recall on answer-containing sentences, meaning it successfully identifies the most relevant passage sentence in nearly 7 of 10 cases. The progressive disclosure mechanism ensures pedagogically appropriate scaffolding.

9. Limitations & Future Work

Limitations:

1. **Model A accuracy ceiling:** Classical TF-IDF features cannot capture long-range semantic relationships. BERT-based models achieve 72% on RACE vs our 35.73% multiple-choice accuracy.

2. **Distractor quality:** NP chunking produces short, sometimes incoherent distractors for abstract questions. The system works best for factual, entity-focused questions.
3. **Question generation:** The rule-based wh-question generator using POS patterns produces grammatically simple questions. Complex inference-based questions from RACE cannot be replicated without generative language models.
4. **Custom article limitations:** The question generator may fail to find a suitable (subject, verb, object) pattern in all sentences, falling back to generic questions.
5. **Evaluation scope:** BLEU/ROUGE/METEOR evaluate surface overlap, not plausibility or educational value. Human evaluation would provide a more meaningful assessment.

Future Work:

- Replace TF-IDF with sentence-transformer embeddings (e.g., all-MiniLM-L6-v2) for richer feature representations.
- Integrate a fine-tuned T5 or GPT-2 model for question generation to produce inference-level questions.
- Add difficulty estimation to select hint sentences at appropriate cognitive levels.
- Implement a user study to evaluate quiz quality, hint usefulness, and learner satisfaction.

10. Conclusion

This project demonstrates that a classical NLP pipeline — without transformer-based models — can produce functional, evaluable reading comprehension quizzes from arbitrary English passages. The system integrates a binary classification pipeline (Model A) for answer verification with an extractive distractor and hint generation system (Model B), all exposed through a four-screen Streamlit interface. Model A achieves 35.73% true multiple-choice accuracy (+10.73% over random) on the RACE dataset. Model B generates lexically diverse, syntactically consistent distractors evaluated at BLEU 0.011, ROUGE-L 0.061, and METEOR 0.025. The hint generation LR scorer achieves 82% accuracy in identifying high-relevance sentences. The project establishes a strong classical ML baseline and a clear upgrade path toward transformer-based methods.

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